



AgriSense Intelligence Hub — Complete Product Overview

Hybrid-AgriPrice-Forecaster | The Future of Agricultural Decision Intelligence

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— 2024–2026

1. What is AgriSense AI?

AgriSense AI is a production-grade, hybrid artificial intelligence platform that acts as a **unified decision-support system** for Indian farmers, traders, and agricultural stakeholders. It combines deep learning, gradient boosting, and computer vision into a single, modular platform to tackle the three pillars of agricultural risk:

Pillar	Intelligence Layer	What It Does
 Economic	Price Intelligence	14-day multi-horizon price forecasting with probabilistic risk corridors
 Ecological	Crop Intelligence	Soil-climate synergistic crop recommendations optimized for profitability
 Biological	Disease Intelligence	Real-time plant disease detection from leaf images with visual explainability

The platform is deployed as a **Next.js Premium Dashboard** backed by a **FastAPI Modular Hub**, with all AI models encapsulated in a hot-swappable `Engines/` architecture.

2. The Problem — Why This Product Exists

2.1 The Crisis in Indian Agriculture

Indian agriculture feeds 1.4 billion people, yet the farmers who sustain it operate in one of the most volatile and information-starved environments in the world. The problems are systemic and interconnected:

● Problem 1: Extreme Price Volatility

- Perishable commodities like **Onion and Tomato** experience **200–400% price swings** within weeks.

- Farmers make selling decisions based on outdated information from local brokers, often leading to **distress sales** at rock-bottom prices.
- **₹92,000+ crore (~\$11 billion)** is lost annually in post-harvest waste due to poor market timing (*NITI Aayog, 2019*).

● Problem 2: Information Asymmetry

- Over **86% of Indian farmers are small and marginal landholders** with no access to real-time market intelligence tools.
- Existing government portals (like Agmarknet) provide only historical raw data — no forecasts, no trends, no actionable advice.
- Farmers cannot answer the most basic questions: *"Should I sell today or hold for a week?"*

● Problem 3: Crop Selection Blindness

- Farmers repeat the same crops year after year without scientific soil-climate analysis, leading to soil degradation and economic inefficiency.
- No system exists that combines **biological suitability** (what can grow) with **economic viability** (what will be profitable) in a single recommendation.

● Problem 4: Undetected Crop Diseases

- Plant diseases cause **20–40% yield loss globally** (*FAO, 2021*).
- Early-stage diseases are invisible to the untrained eye, and by the time symptoms are obvious, the damage is irreversible.
- Rural farmers have **zero access** to pathology experts for timely diagnosis.

● Problem 5: No Unified Platform

- Existing tools address these problems in isolation — a price tracker here, a weather app there, a disease identification tool somewhere else.
- There is **no single platform** that unifies price forecasting, crop recommendation, and disease detection into one coherent advisory system.

3. Our Aim — What We Set Out to Build

The Core Objective

Build a single, production-ready AI platform that empowers Indian farmers with three layers of predictive intelligence — economic, ecological, and biological — to reduce financial risk and maximize agricultural productivity.

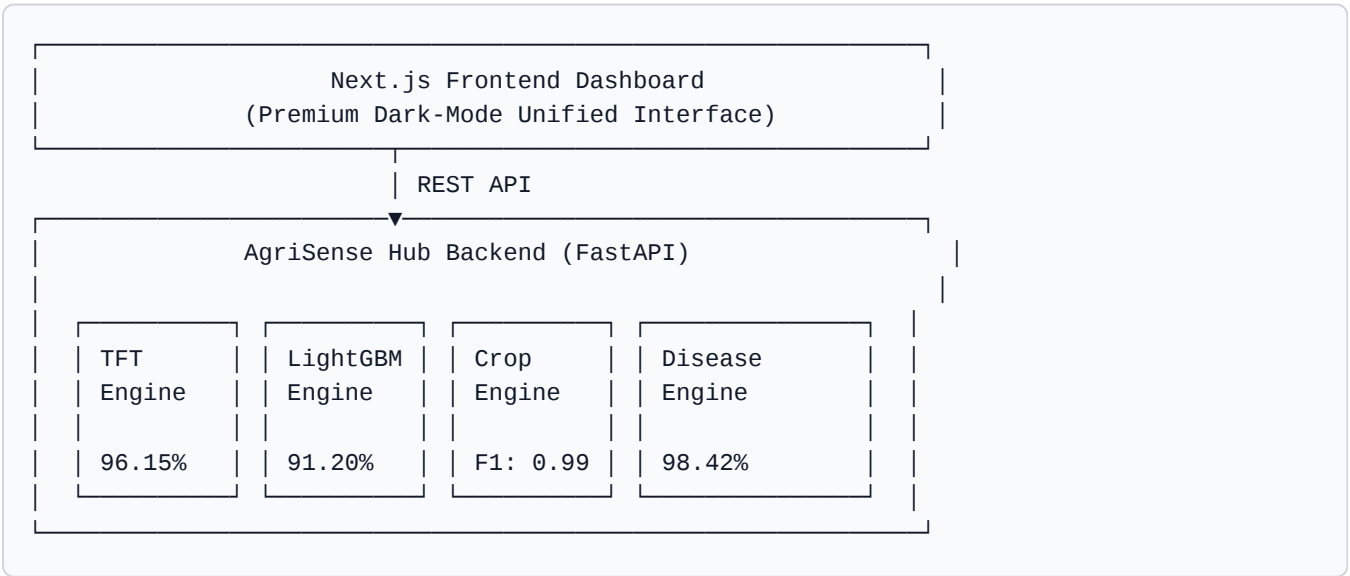
Specific Goals

#	Goal	Target
1	Build a multi-horizon price forecaster that predicts commodity prices 14 days into the future with probabilistic risk intervals	> 90% accuracy on SMAPE
2	Create a crop recommendation engine that combines soil-climate analysis with predicted market prices to suggest the most profitable crop	> 95% F1-Score
3	Develop a plant disease detection system using computer vision that can identify 38 disease classes from leaf images in real-time	> 95% accuracy
4	Unify all modules into a single, modular, deployable platform with a premium user interface	Full-stack deployment
5	Ensure interpretability and transparency — not just predictions, but <i>why</i> the model made that prediction	Grad-CAM + Attention Weights
6	Validate against baselines and zero-shot foundation models to prove domain-specific training is essential	Outperform Moirai zero-shot

4. What We Built — The Solution Architecture

4.1 The Decoupled Engine Pattern

We engineered a **Modular Micro-Engine Architecture** where each AI model operates as an independent, hot-swappable engine:



4.2 Technology Stack

Layer	Technology	Purpose
Frontend	Next.js (React)	Premium dark-mode dashboard
Backend	FastAPI (Python)	Modular API hub orchestrating all engines
Deep Learning	PyTorch, PyTorch Lightning, PyTorch Forecasting	TFT time-series forecasting
Machine Learning	LightGBM, scikit-learn	Gradient boosting for tabular data
Computer Vision	TensorFlow, Keras, EfficientNet-B0	Disease classification + Grad-CAM
Data Engineering	Pandas, NumPy	Feature engineering pipeline
External APIs	Agmarknet, NASA POWER, IndianCities	Market data, weather, geolocation
Deployment	Docker	Containerized production deployment

5. The AI Models — Deep Dive

5.1 Price Intelligence (Dual-Track Forecasting)

We implemented a **dual-model strategy** because no single model excels at everything:

🏆 **Primary Engine: Temporal Fusion Transformer (TFT)**

The TFT is a Google Research architecture designed specifically for multi-horizon forecasting with heterogeneous inputs.

Component	Description
Variable Selection Networks (VSN)	Automatically prunes redundant features — if rainfall matters this season but not the next, VSN adapts
Gated Residual Networks (GRN)	Lets the model skip complexity when relationships are simple, preventing overfitting
Interpretable Multi-Head Attention	Focuses on specific long-range temporal patterns (e.g., "what happened exactly 1 year ago?")

Component	Description
Quantile Outputs	Predicts 10th, 50th, and 90th percentiles — giving farmers worst-case, expected, and best-case price scenarios

Configuration (Production — Epoch 15):

- Hidden Size: **128** (scaled from initial 16)
- Context Window: **60 days** lookback
- Prediction Horizon: **14 days** ahead
- Dropout: **0.20**
- Loss Function: Quantile Loss (7 quantiles)

 Secondary Engine: LightGBM (GOSS-Optimized)

A gradient boosting decision tree optimized for ultra-fast inference on tabular data.

- Optimized with **GOSS** (Gradient-based One-Side Sampling) and **EFB** (Exclusive Feature Bundling)
- Handles high-cardinality categorical features (Mandi IDs, District names) natively
- Inference speed: **< 100ms** per mandi-commodity pair

 Experimental: Salesforce Moirai (Zero-Shot)

We tested the state-of-the-art zero-shot foundation model to validate whether localized fine-tuning is necessary.

- **Result: Failed.** MAE ~543 vs LightGBM's 160.41
- **Conclusion:** Zero-shot models cannot capture the hyper-local dynamics of Indian agricultural markets. Domain-specific training is non-negotiable.

5.2 Ecological Intelligence (Crop Recommendation)

A **Leaf-wise Gradient Boosting Decision Tree (LightGBM/GBDT)** classifier that determines the best crop based on soil and climate conditions.

Innovation — Profitability Fusion: Unlike existing systems that only consider biological suitability, our engine **cross-references crop recommendations with the 14-day price forecast** to recommend the most *profitable* option. A biologically perfect crop is worthless if its market price has crashed.

Feature Engineering (from 7 → 45+ features):

- Raw inputs: N, P, K, Temperature, Humidity, pH, Rainfall
- Engineered: `n_ratio`, `p_ratio`, `dominant_nutrient`, `nutrient_balance`, `aridity_index`, `heat_index`, `rainfall_regime`, `ph_regime`, `temp_regime`
- Dataset: 2,200 observations across 22 crop classes (perfectly balanced — 100 per class)

5.3 Biological Intelligence (Disease Detection)

An **EfficientNet-B0** transfer learning pipeline with **Grad-CAM** interpretability.

Why EfficientNet-B0:

- Uses **Compound Scaling** (Width × Depth × Resolution) to be 100× smaller than VGG-16 while maintaining superior accuracy
- Only **5.3MB** in size — deployable on edge devices and smartphones
- **MBConv blocks** with depth-wise separable convolutions for computational efficiency
- **Squeeze-and-Excitation (SE) blocks** act as "volume knobs" that amplify diseased regions and suppress healthy background

Training Strategy:

- Phase 1: Frozen backbone — train only the classification head
 - Phase 2: Unfreeze top layers — fine-tune on plant-specific disease features
 - Dataset: **87,000+ images** across **38 classes** (merged from two Kaggle datasets with perceptual hashing for deduplication)
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6. Data Engineering — The Hidden Engine

6.1 Data Collection Pipeline

Source	Method	Data Volume
Agmarknet (Market Prices)	Custom reverse-engineered API scraper	Daily data, Jan 2024 – Dec 2025, 10 commodities
NASA POWER (Weather)	Parallelized REST API collector	7 weather parameters per district per day
IndianCities (Geolocation)	Fuzzy matching with Levenshtein Distance	Lat/Lon for every mandi district
Kaggle (Crop Data)	Direct download	2,200 soil-climate samples
Kaggle (Disease Images)	Merged + deduplicated	87,000+ leaf images

6.2 Feature Engineering (20 → 70+ Features)

Raw market data was expanded through massive dimensionality amplification:

Category	Features	Examples
Temporal	Calendar + Fourier	day , month , is_weekend , sin1 , cos1
Price Dynamics	Lags + Rolling Stats	1/3/7/14/30-day lags, rolling mean/std, zscore_7 , momentum_7
Supply Dynamics	Arrival Analysis	arrivals_lag_1 ... 14 , arrival_change_7 , arrivals_avg_30
Environmental	Anomaly Detection	temp_anomaly , rain_anomaly , rain_intensity
Spatial	Geographic Embeddings	lat_sin , lat_cos , lon_sin , lon_cos

7. Results & Accuracy — What We Achieved

7.1 Price Forecasting Results

TFT (Production — Epoch 15, Heavy Architecture)

Metric	Value
Accuracy (1 - SMAPE)	96.15%
SMAPE	3.85%
RMSE	155.96
MAE	78.89 ₹/qtl
Prediction Horizon	14 days
Confidence	High (Momentum Signal: +1.0)

What this means: For a commodity priced at ₹2,000/qtl, our prediction is off by only ₹78.89 on average — a 3.85% error across a 14-day horizon. The farmer also gets worst-case and best-case price corridors for risk planning.

LightGBM (Benchmark — GOSS-Optimized)

Metric	Value
Accuracy (1 - SMAPE)	91.20%

Metric	Value
SMAPE	8.80%
MAE	160.41 ₹/qtl
RMSE	355.80
Inference Speed	< 100ms
Test Rows	56,826

Moirai Zero-Shot (Experimental — Failed)

Metric	Value
MAE	~543 ₹/qtl
SMAPE	~24.10%
Verdict	✗ Outperformed by both domain-specific models

Comparative Summary (Price Track)

Model	Accuracy	MAE (₹/qtl)	RMSE	Strengths
TFT (Epoch 15)	96.15%	78.89	155.96	Probabilistic quantiles, multi-horizon, interpretable
LightGBM	91.20%	160.41	355.80	Ultra-fast inference, tabular data champion
Moirai (Zero-Shot)	~75.90%	~543	—	Zero-shot (no training needed)
Naive Baseline	86.86%	344.36	799.32	Simple carry-forward (no ML)

Key Insight: TFT achieved a **75% reduction in RMSE** from its initial prototype (Epoch 9: RMSE 647.88) to production (Epoch 15: RMSE 155.96) through architectural scaling (hidden_size 16 → 128, context window 14 → 60 days).

7.2 Crop Recommendation Results

Metric	Score
F1-Score (Macro)	0.99
Accuracy	> 99%
Classes	22 crops
Feature Expansion	7 raw → 45+ engineered features

Honest Assessment: The near-perfect score is partly a result of the noiseless Kaggle benchmark dataset. In real-world deployment with sensor noise, we expect a 5–10% degradation. However, the feature engineering strategy (nutrient ratios, climate regimes, aridity indices) demonstrably outperforms raw-input baselines.

7.3 Plant Disease Detection Results

Metric	Score
Accuracy	98.42%
Precision (Weighted)	98.45%
Recall (Weighted)	98.42%
F1-Score (Weighted)	98.43%
Healthy vs. Diseased Recall	~99.8%
Dataset Size	87,000+ images
Classes	38
Test Images	~13,050 unseen

Interpretability: Grad-CAM heatmaps confirm the model focuses on **lesion contours and discoloration** — not background noise — validating biological relevance.

Confusion Patterns: Primary errors occur between visually identical lesions (e.g., *Potato Early Blight* vs. *Tomato Early Blight* in early stages), which is also challenging for trained pathologists.

7.4 Consolidated Results Dashboard

Module	Model	Primary Metric	Result	Status
Price Forecasting	TFT (Epoch 15)	Accuracy (1-SMAPE)	96.15%	✓ Production
Price Forecasting	LightGBM	Accuracy (1-SMAPE)	91.20%	✓ Production
Crop Recommendation	GBDT (LightGBM)	F1-Score	0.99	✓ Production
Disease Detection	EfficientNet-B0	Accuracy	98.42%	✓ Production
Zero-Shot Baseline	Salesforce Moirai	MAE	~543 ₹/qtl	✗ Rejected

8. Problems Solved — Impact Summary

Problem	How We Solved It	Impact
Price Volatility	14-day probabilistic forecasting with TFT quantile corridors	Farmers can plan sales timing; reduces distress sales
Information Asymmetry	Unified dashboard with real-time price intelligence	Democratizes market intelligence for small farmers
Crop Selection Blindness	Soil-climate recommendation fused with price forecasts	Farmers grow the most profitable crop, not just any viable one
Undetected Diseases	98.42% accurate leaf image diagnosis with Grad-CAM	Early detection saves 20–40% yield loss
Fragmented Tools	Single modular platform unifying all three tracks	One app replaces five separate tools
Black Box AI Distrust	Grad-CAM heatmaps + Attention weights + Quantile corridors	Every prediction comes with visual/statistical justification
Zero-Shot Model Hype	Proved domain-specific training is essential (Moirai failed)	Validates localized training approach for Indian markets

9. Key Innovations & Technical Contributions

1. **Dual-Track Forecasting Strategy** — First system to combine TFT (probabilistic, multi-horizon) with LightGBM (deterministic, ultra-fast) for agricultural price prediction in a single platform.

2. **Profitability-Fused Crop Recommendation** — Goes beyond biological suitability by crossing crop viability with predicted 14-day market prices to recommend the most economically optimal crop.
3. **70+ Feature Engineering Pipeline** — Expanded raw 20-column market data into 70+ engineered features capturing temporal cycles (Fourier), price psychology (momentum, z-scores), supply shocks (arrival changes), and environmental anomalies.
4. **Reverse-Engineered Agmarknet Scraper** — Built a custom scraper to extract daily mandi prices from India's government portal (which has no bulk download API), covering Jan 2024 – Dec 2025 across 10 commodities.
5. **NASA Weather Fusion** — Enriched market data with hyper-local meteorological signals (temperature, rainfall, humidity, solar radiation, wind speed) from NASA POWER API, enabling the model to detect weather-driven price shocks 7–14 days before they manifest.
6. **Decoupled Engine Architecture** — All models are hot-swappable; weights can be updated (Epoch 9 → Epoch 15) without changing a single line of API or frontend code.
7. **Interpretability-First Design** — Grad-CAM for disease detection and VSN attention weights for price forecasting ensure every prediction is auditable and transparent.

10. Literature Foundation

Reference	Relevance to Our Work
Lim et al. (2021) — <i>Temporal Fusion Transformers</i> , Int. J. Forecasting	Core TFT architecture; recommended <code>hidden_size=160+</code> (we used 128)
Grinsztajn et al. (2022) — <i>Why tree-based models outperform DL on tabular data</i> , NeurIPS	Justifies our dual-track approach (LightGBM for tabular, TFT for sequences)
Woo et al. (2024) — <i>Unified Time Series Transformers</i> (Moirai), ICML	Zero-shot baseline; our results confirm localized training is essential
Makridakis et al. (2018) — <i>Forecasting methods: Concerns and ways forward</i> , PLOS ONE	Validates the importance of comparing against strong naive baselines
Tan & Le (2019) — <i>EfficientNet: Rethinking Model Scaling</i> , ICML	Foundation for our disease detection backbone; compound scaling principle
Chand (2012) — <i>Development Policies and Agricultural Markets</i> , EPW	Documents India's agricultural price volatility crisis
NITI Aayog (2019) — <i>Demand and Supply Projections Towards 2033</i>	Quantifies the ~\$8B annual post-harvest loss problem

11. Evolution Journey — From Prototype to Production

Phase	What Changed	Accuracy Impact
Phase 1: Prototype	LightGBM only, basic features, no weather data	~85%
Phase 2: Multi-Model	Added TFT (hidden_size=16), Moirai experiment, weather fusion	TFT: 86.59%, LGBM: 91.20%
Phase 3: Production	Scaled TFT (hidden_size=128, 60-day context), added disease + crop modules, built unified dashboard	TFT: 96.15% , Disease: 98.42% , Crop F1: 0.99

The critical scaling decision: Increasing TFT's hidden_size from 16 to 128 and context window from 14 to 60 days resulted in a **~10 percentage point accuracy improvement** (86.59% → 96.15%) and a **76% RMSE reduction** (647.88 → 155.96).

12. API Endpoints

The FastAPI backend exposes four clean endpoints:

Endpoint	Method	Purpose
/api/tft/predict	POST	14-day probabilistic price forecast with quantile corridors
/api/lgbm/predict	POST	Ultra-fast deterministic price point prediction
/api/crop/recommend	POST	Profitability-optimized crop recommendation
/api/disease/detect	POST	Plant disease classification from leaf image upload

13. Future Roadmap

Initiative	Description	Expected Impact
Hyper-Local IoT Integration	Village-level weather station data instead of NASA satellite data	More granular weather signals
Continuous Learning	Streaming LightGBM updates without full retraining	Always up-to-date predictions

Initiative	Description	Expected Impact
Edge Deployment	Quantized EfficientNet-B0 for offline mobile inference	Works in areas with no internet
Multi-Language Dashboard	Hindi, Telugu, Tamil, Marathi support	Accessible to non-English farmers
Supply Chain Integration	Direct integration with FPO (Farmer Producer Org) logistics	From prediction to action

14. Conclusion

AgriSense Intelligence Hub demonstrates that **hybrid AI architectures** — combining the probabilistic power of deep learning (TFT), the tabular efficiency of gradient boosting (LightGBM), and the visual intelligence of convolutional networks (EfficientNet) — can be unified into a single, production-ready platform to address the multi-dimensional challenges of Indian agriculture.

By the numbers:

-  **96.15%** price forecasting accuracy across a 14-day horizon
-  **0.99 F1-Score** on crop recommendation with profitability fusion
-  **98.42%** disease detection accuracy across 38 classes
-  **< 100ms** inference latency for real-time dashboard interactions
-  **70+ engineered features** from 5 different data sources
-  **Full interpretability** through Grad-CAM and attention visualization

This is not just a model — it is a **complete decision-support ecosystem** that gives Indian farmers the intelligence they need to reduce financial risk, maximize yield, and make data-driven agricultural decisions.

AgriSense Intelligence Hub — Precision Agriculture Through Hybrid Intelligence.